

SYNOPSIS

AI-Based Intelligent Chatbot for Academic and Administrative Support on College Website

Codec-Technologies – Artificial Intelligence Internship

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Introduction

- 1 College websites host vast academic & administrative information, but content is scattered across pages — making quick access difficult.
- 2 Users depend on emails, calls, or in-person visits, creating delays and extra administrative workload.
- 3 AI-based conversational systems (ChatGPT, Gemini, etc.) have shifted user behavior toward natural, chat-based information retrieval.
- 4 An institution offering AI/ML (with IBM collaboration) should reflect this innovation on its own website.

Why a Chatbot?

24/7

Availability, no office-hour limits

Instant

Real-time query resolution

Consistent

Accurate, standardized answers

Reduced Load

Less repetitive admin workload

Problem Statement

Scattered content, complex navigation, and no interactive support prevent users from quickly finding what they need on the college website — driving repeated queries to staff and reducing overall user experience.



Fragmented Information

Content spread across many pages & documents, requiring manual, repeated searches.



Administrative Burden

Routine queries flood email, phone, and in-person channels, consuming staff time.



No Conversational Access

No AI-driven interface exists, unlike the tools users expect from modern platforms.

Objectives of the Project

1

Identify common information-accessibility issues on the college website

2

Design an AI chatbot that understands queries using NLP techniques

3

Deliver instant, accurate, consistent answers to FAQs

4

Reduce repetitive workload on administrative offices

5

Enable 24/7 conversational support to boost engagement

6

Demonstrate applied AI/ML aligned with the B.Tech curriculum

7

Build a scalable solution ready for institutional integration

Literature Review — Comparative Analysis

System	Technique	Strengths	Limitations
College Enquiry Bot	NLP + Q&A	Simple, efficient	Limited flexibility
AI Student Bot	NLP + ML	Better intent recognition	Requires frequent updates
RAG Bot	Retrieval + Generation	Context-aware	High complexity
Rule-Based Bot	Rule logic	Predictable	Poor adaptability



Research Gap

There is a lack of moderately intelligent, institution-specific chatbot systems that balance feasibility, accuracy, and ease of deployment while remaining suitable for undergraduate-level implementation.

Trends from the Literature



Benefits Identified

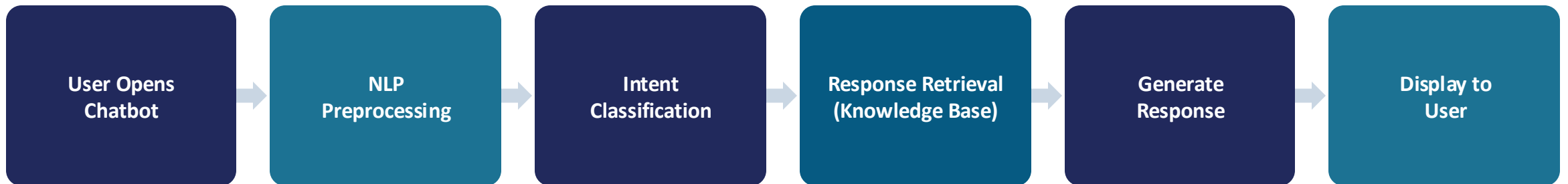
- Instant support with no waiting time
- Enhanced accessibility, anytime access
- Improved student engagement
- Reduced administrative workload
- Personalized, context-aware assistance



Challenges Identified

- Limited context understanding
- Heavy dependency on quality data
- Ethical & accuracy concerns
- Website integration complexity
- Frequent dataset/model updates needed

Methodology — System Flow



A retrieval-based pipeline: user input is cleaned by NLP preprocessing, classified into an intent, matched against a structured knowledge base, then formatted into a response for the user.

NLP Preprocessing

Tokenization, lemmatization, stop-word removal

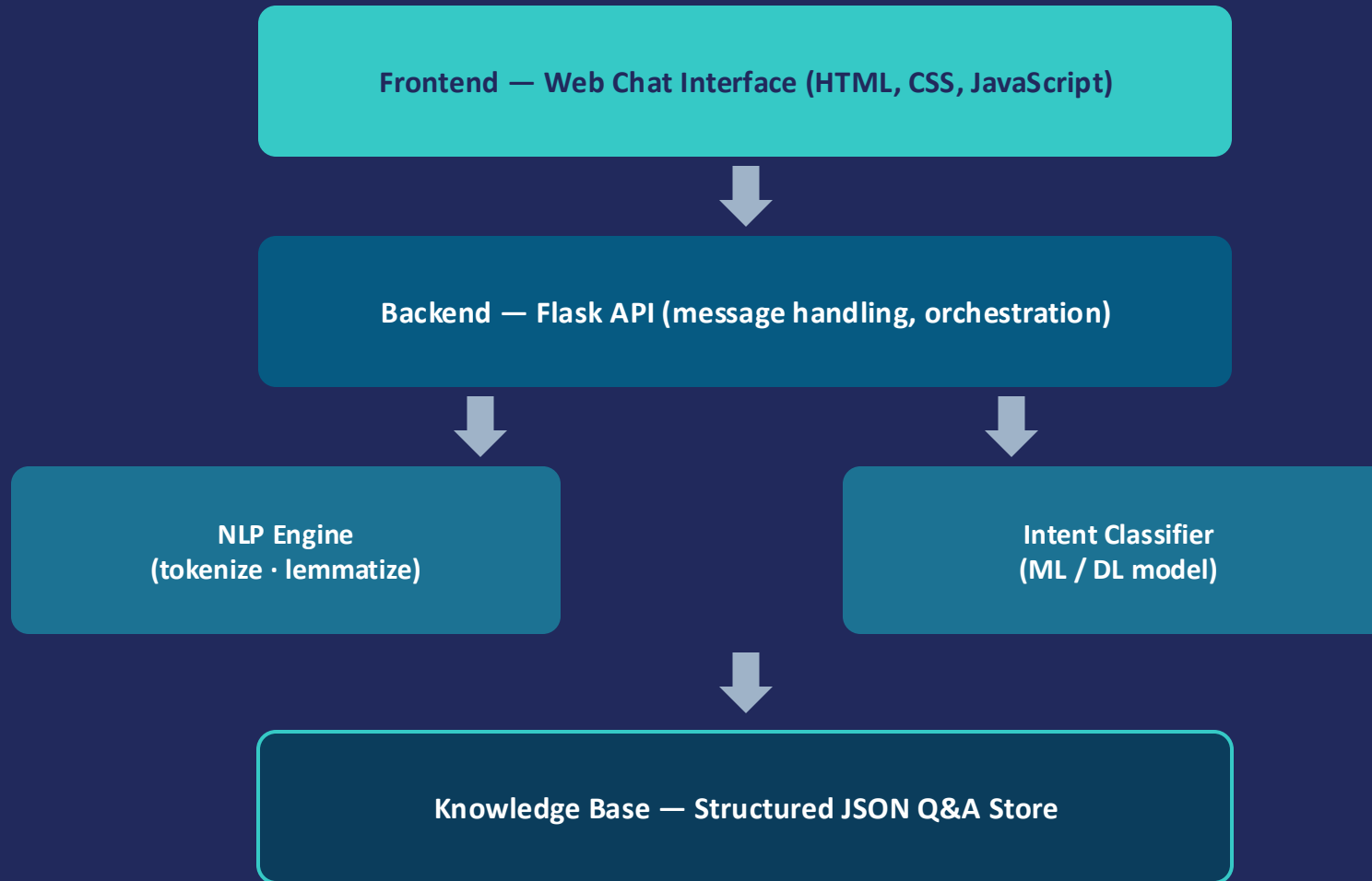
Intent Classification

ML/DL model predicts query category

Knowledge Base

Structured JSON of verified Q&A pairs

System Architecture



Modular 3-tier design: presentation, processing, and data layers communicate through a lightweight REST API, keeping responses aligned with verified institutional information.

Software & Hardware Requirements

SW

Software Stack

- Python — core development language
- Flask — backend REST API server
- TensorFlow / Keras — deep learning model
- Scikit-learn — ML classification models
- NLTK — NLP preprocessing toolkit
- HTML, CSS, JavaScript — chat UI

HW

Hardware Requirements

- Minimum 8GB RAM
- Intel i5 or equivalent processor
- Stable internet connectivity

A lightweight, retrieval-based design keeps compute requirements modest — suitable for a standard development machine.

Project Timeline — 13 Weeks

Wk 1

Problem Identification

Wk 2

Literature Review

Wk 3

Requirement Analysis

Wk 4

System Design

Wk 5-6

Data Collection & Structuring

Wk 7

Data Preprocessing

Wk 8

Model Development

Wk 9

Backend Development

Wk 10

Frontend Development

Wk 11

Integration & Testing

Wk 12

Optimization

Wk 13

Documentation

Future Work

1

Expand FAQ Database

Continuously add new questions & updated info to widen coverage.

2

Multilingual Support

Extend to Hindi and other languages via separate intent maps.

3

Admin Dashboard

Flask-based panel for staff to update responses without touching code.

4

Query Analytics

Log & analyze queries to spot gaps and improvement areas.

5

Improved Intent Handling

Periodically retrain the model on real user queries.

Conclusion

The AI-powered college website chatbot tackles scattered information and delayed responses with an intelligent, automated conversational interface — improving accessibility, cutting administrative workload, and showcasing practical AI/NLP application.

Accessible

Scalable

Institution-Ready

AI-Driven

Thank You

Questions & Discussion Welcome